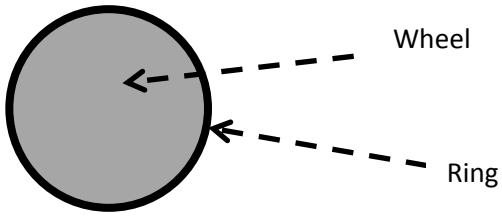
Multiple choise answers:

Question 1	a	**b**	c	d	e	f
Question 2	**a**	b	c	d	e	f
Question 3	a	b	c	**d**	e	f
Question 4	a	**b**	c	d	e	f
Question 5	a	b	c	**d**	e	f
Question 6	a	**b**	c	d	e	f
Question 7	**a**	b	c	d	e	f
Question 8	a	b	c	**d**	e	f
Question 9	**a**	b	c	d	e	f
Question 10	a	**b**	c	d	e	f
Question 11	**a**	b	c	d	e	f
Question 12	a	b	**c**	d	e	f
Question 13	**a**	b	c	d	e	f
Question 14	**a**	b	c	d	e	f
Question 15	a	b	c	**d**	e	f
Question 16	a	b	**c**	d	e	f
Question 17	a	b	c	**d**	e	f
Question 18	a	**b**	c	d	e	f
Question 19	a	b	**c**	d	e	f
Question 20	a	**b**	c	d	e	f
Question 21	a	b	c	**d**	e	f
Question 22	a	b	c	**d**	e	f
Question 23	a	**b**	c	d	e	f
Question 24	a	b	**c**	d	e	f

Problem 1 B

.

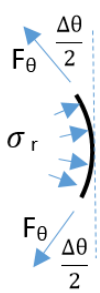


Let σ_r and F_θ be the radial pressure on the contact surface and the force in the ring respectively. We have:

$$\Delta l = l\alpha\Delta T = 2\pi r\alpha\Delta T$$

$$F_\theta = k\ \Delta l = k\ 2\pi r\alpha\Delta T$$

Let t be the thickness of the ring. From the force equilibrium for a very small section of the ring (Δθ) we have:



$$2\ F_\theta\ sin\left(\frac{\Delta\theta}{2}\right) = \ 2\ F_\theta\ \left(\frac{\Delta\theta}{2}\right) = (\Delta S)\ \sigma_r = (tr\Delta\theta)\ \sigma_r$$

$$(tr)\ \sigma_r = F_\theta\ =\ k\ 2\pi r\alpha\Delta T$$

$$`\sigma_r = \frac{k\ 2\pi\alpha\Delta T}{t}$$

The force required for pushing the ring out F is:

$$F = \mu N = \mu 2\pi rt\ \sigma_r$$

$$F = 4\pi^2\mu kr\alpha\Delta T$$

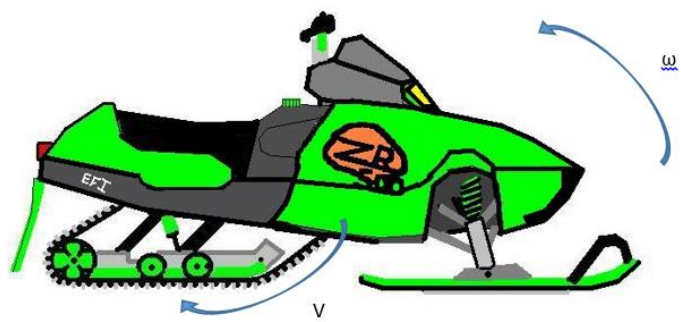
$$F = 4\pi^2(0.8)(2.1\times 10^7)(1.2\times 10^{-5})(0.375)(400-23)\ N = \ 1.13\times 10^6\ N$$

Problem 2 B.

Gravitational force is the only force that acts on the system of the snowmobile and the rider in the air, therefore the total external torque around the center of the mass of the system is zero and the angular momentum around this point should remain constant.

Since the weight of the rotating track is comparable to the weight of the system, any change in its rotational speed results in a rotation of the snowmobile in the opposite direction. So if the rider accelerates the front of the snowmobile will move upward and if the person brakes it will move down.

In this question, the rider should decelerate to tilt the head down and accelerate back to the previous speed to keep it at that angle. Since there is no drag force on the track, we assume that any deceleration and acceleration can happen almost instantly (compared to the jump time).



From the pictures we approximate the radius of gyration to be 1 m for the system of the rider and the snowmobile and 0.5 m for the track.

$$T_{before\ the\ jump} = T_{after\ the\ jump}$$

$$m_{track}\, r_{track}\, v - m_{driver+snowmobile}\, (r_{driver+snowmobile})^2 \omega = m_{track}\, r_{track}\, v_0$$

Where V_0 is the velocity of rotation of the snowmobile before braking.

After applying the brakes we have:

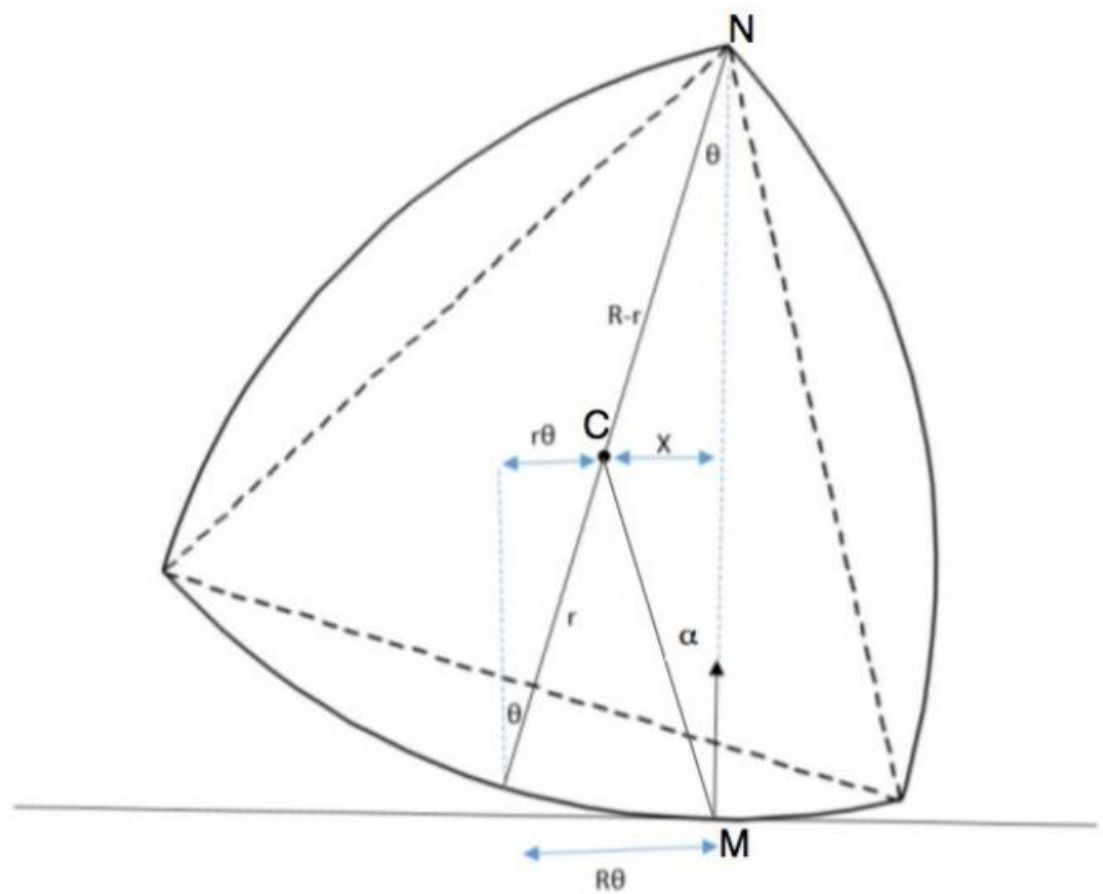
$$\omega = \frac{m_{track}\, r_{track}\, (v - v_0)}{m_{driver+snowmobile}\, {r_{snowmobile}}^2} \approx (0.04 \frac{rad}{m})(\Delta v)$$

Assuming that the jump time is around 2 seconds, the desired ω should be 2.5°/s

$$\Delta v \approx \frac{\omega}{0.04} = \frac{\frac{\delta\theta}{t}}{0.04} = \frac{\frac{0.0873}{2}}{0.04} \approx 1.1 \frac{m}{s} \approx \frac{4.0km}{h}$$

Problem 3 B.

Method 1



$Writing\ Newton's\ law\ for\ the\ horizontal\ and\ vertical\ components\ of\ the\ force\ for\ a\ small\ \theta$:

$$\Sigma F_y = N - mg = ma_y \approx 0$$

$$F_{friction} = ma_x \approx mr\ddot{\theta}$$

$$a_y \approx 0\,, a_x \approx r\ddot{\theta}$$

$$\rightarrow N \approx mg, F_{friction} \approx mr\ddot{\theta}$$

$The\ torque\ equation\ about\ the\ point\ C, gives$:

$$mI\ddot{\theta} = Nx + F_{friction}r$$

$Substituting\ for\ F_{friction}\ and\ N, we\ get$:

$$(mI + mr^2)\ddot{\theta} = mgx$$

$On\ the\ other\ hand$:

$$x = (R - r)\sin\theta \approx (R - r)\,\theta = \frac{2}{3}\,R\sin\left(\frac{\pi}{3}\right)\theta$$

$$R - r = \frac{2}{3}\,R\sin\left(\frac{\pi}{3}\right) =$$

$$\rightarrow\ r = R(1 - \frac{\sqrt{3}}{3})$$

$$\rightarrow\ r^2 = \frac{4 - 2\sqrt{3}}{3}R^2$$

$Substituting\ for\ r^2 and\ x\ in\ the\ equation\ for\ \ddot{\Theta}$

$$\left(CR^2 + \frac{4 - 2\sqrt{3}}{3}R^2\right)\ddot{\Theta} = g\frac{R\sqrt{3}}{3}\theta$$

$Which\ result\ in\ a\ harmonic\ oscillator\ with$:

$$\omega^2 = \frac{g}{\sqrt{3}\,R(C + \frac{4 - 2\sqrt{3}}{3})} = \frac{g\sqrt{3}}{R(3C + 4 - 2\sqrt{3})}$$

METHOD 2:

Alternatively, using the Energy we could write:

$$y_{cm} = y_N - (R - r)\,cos\theta = R - (R - r)cos\theta$$

$$x_{cm} = x_N - (R - r)\,sin\theta = R\theta - (R - r)sin\theta$$

$$v_{cm}^{x} = (R - r)sin\theta\,\dot{\theta}$$

$$v_{cm}^{y} = [R - (R \dot{-} r)cos\theta]\,\dot{\theta}$$

$$E = \frac{m}{2}\left(V_{cm}^{x\ 2} + V_{cm}^{y\ 2}\right) + \frac{I_{cm}}{2}\dot{\theta}^2 + mgy_{cm}$$

$$\rightarrow E = const = \frac{m}{2}(R^2 + (R - r)^2 - 2R(R - r)cos\theta)\dot{\theta}^2 + \frac{mCR^2}{2}\dot{\theta}^2 + mg[R - (R - r)cos\theta]$$

For a small angle:

$$E \approx \frac{m}{2}\left(R^2 + (R - r)^2 - 2R(R - r)\right)\dot{\theta}^2 + \frac{mCR^2}{2}\dot{\theta}^2 + mg\left[R - (R - r)(1 - \frac{\theta^2}{2})\right]$$

Since E is constant and $\dot{E} = 0$:

$$\frac{m}{2}(r^2)2\dot{\theta}\ddot{\theta} + \frac{mCR^2}{2}\,2\dot{\theta}\ddot{\theta} + mg(R-r)\dot{\theta}\theta = 0$$

$$\rightarrow m(r^2 + CR^2)\ddot{\theta} + mg(R-r)\theta = 0$$

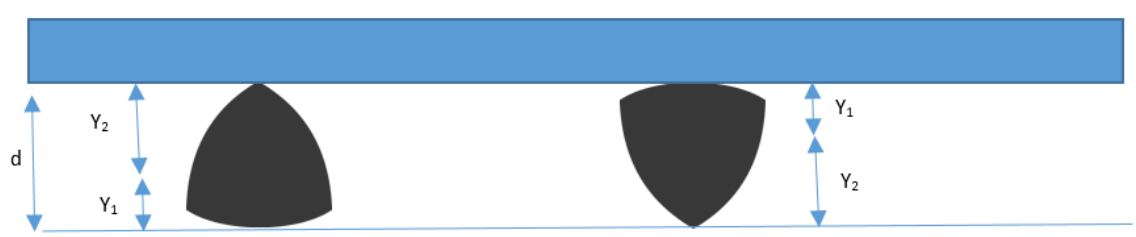
Substituting for r from method 1 ,results in the same equation:

$$\left(CR^2 + \frac{4-2\sqrt{3}}{3}R^2\right)\ddot{\Theta} = -g\frac{R\sqrt{3}}{3}\theta$$

Which is a harmonic oscillator with

$$\omega^2 = \frac{g}{\sqrt{3}\,R(C + \frac{4-2\sqrt{3}}{3})} = \frac{g\sqrt{3}}{R(3C + 4 - 2\sqrt{3})}$$

a) The distance between the plank and the ground is always constant and equal to the radius of the Reuleaux triangle.

b) If we place the logs in such a way that log 1 is always 180° rotated compared to log 2 the sum of the height of the center of the masses of the two logs will remain constant(figure below). Also from part b we know that the height of the plank is always constant. Therefore the height of the center of the masses of the system remains constant and the plank will roll smoothly.

$$center\ of\ mass\ of\ the\ logs = \frac{Y_1 + Y_2}{2} = \frac{d}{2} = constant$$